Susceptibility of Fermi gas and 2-particle GF

$$\chi_0(\vec{r},t) = -i\theta(t)\left\langle [\hat{\rho}(\vec{r},t), \hat{\rho}(0,0)]\right\rangle_{H_0}$$

$$\hat{\rho}(\vec{r}) = \hat{\psi}^+(r)\hat{\psi}(r) = \frac{1}{V}\sum_{kk'} \hat{a}^+_k \hat{a}_{k'} e^{-i(\vec{k}-\vec{k}')\vec{r}}$$

$$\hat{H}_0 = \sum_k \left(\frac{\hbar^2 k^2}{2m} - \mu\right) a^+_k a_k$$

$$\hat{\rho}(\vec{r},t) = e^{i\hat{H}_0 t/\hbar}\,\hat{\rho}(\vec{r})\,e^{-i\hat{H}_0 t/\hbar} =$$

$$= \frac{1}{V}\sum_{kk'} e^{-i(\vec{k}-\vec{k}')\vec{r}} \underbrace{e^{i\hat{H}_0 t/\hbar}\,\hat{a}^+_k\,e^{-i\hat{H}_0 t/\hbar}}_{\hat{a}^+_k(t)} \underbrace{e^{i\hat{H}_0 t/\hbar}\,\hat{a}_{k'}\,e^{-i\hat{H}_0 t/\hbar}}_{\hat{a}_{k'}(t)}$$

$$i\hbar\,\partial_t \hat{A}(t) = [\hat{A}, \hat{H}]$$

$$\Rightarrow i\hbar\,\partial_t \hat{a}_{k'}(t) = [\hat{a}_{k'}, \hat{H}_0] = \sum_k \left(\frac{\hbar^2 k^2}{2m} - \mu\right)[\hat{a}_{k'}, \hat{a}^+_k a_k]$$

Fermions

$$\underbrace{\hat{a}_{k'}\hat{a}^+_k}_{\delta_{kk'} - \hat{a}^+_k \hat{a}_{k'}}\hat{a}_k - \hat{a}^+_k \hat{a}_k \hat{a}_{k'} = \delta_{kk'}\hat{a}_k - \hat{a}^+_k \underbrace{\hat{a}_{k'}\hat{a}_k}_{-\hat{a}_k\hat{a}_{k'}} - \hat{a}^+\hat{a}_k\hat{a}_{k'}$$

$a_{k'}a^+_k + a^+_k a_{k'} = \delta_{kk'}$

$$= \delta_{kk'}\hat{a}_k$$

$$\Rightarrow i\hbar\,\partial_t \hat{a}_{k'}(t) = \sum_k \delta_{kk'}\left(\frac{\hbar^2 k^2}{2m} - \mu\right)\hat{a}_k(t) =$$

$$= \left(\frac{\hbar^2 k'^2}{2m} - \mu\right)\hat{a}_{k'}(t) = (\varepsilon_{k'} - \mu)\,\hat{a}_{k'}(t)$$

$$\Rightarrow \hat{a}_{k'}(t) = e^{-i(\varepsilon_{k'}-\mu)t/\hbar}\,\hat{a}_{k'}(0)$$

$$\hat{a}_k^+(t) = \left(\hat{a}_k(t)\right)^+ = \left(e^{iH_0 t/\hbar}\, \hat{a}_k\, e^{-iH_0 t/\hbar}\right)^+ =$$

$$= e^{iH_0 t/\hbar}\, \hat{a}_k^+\, e^{-iH_0 t/\hbar}$$

$$\Rightarrow \hat{a}_k^+(t) = \left(e^{-i(\varepsilon_k-\mu)t/\hbar}\, \hat{a}_k(0)\right)^+ = e^{i(\varepsilon_k-\mu)t/\hbar}\, \hat{a}_k^+$$

Note $\hat{a}_{k'}(0) = \hat{a}_{k'}$; $\hat{a}_k^+(0) = \hat{a}_k^+$.

$$\Rightarrow \hat{\rho}(\vec{r},t) = \frac{1}{V} \sum_{kk'} e^{-i(\vec{k}-\vec{k}')\vec{r}}\, e^{i(\varepsilon_k-\mu)t/\hbar}\, e^{-i(\varepsilon_{k'}-\mu)t/\hbar}$$

$$\cdot\, \hat{a}_k^+ \hat{a}_{k'}.$$

$$\chi_0(\vec{r},t) = -i\theta(t)\, \frac{1}{V} \sum_{kk'} e^{-i(\vec{k}-\vec{k}')\vec{r}}\, e^{i(\varepsilon_k-\mu)t/\hbar}\, e^{-i(\varepsilon_{k'}-\mu)t/\hbar}$$

$$\cdot \left\langle \left[\hat{a}_k^+ \hat{a}_{k'}, \underbrace{\hat{\rho}(0,0)}_{\frac{1}{V}\sum_{k_1k_2} \hat{a}_{k_1}^+ \hat{a}_{k_2}}\right]\right\rangle$$

Need to evaluate

$$[a_k^+ a_{k'}, a_{k_1}^+ a_{k_2}] = a_k^+ a_{k'} a_{k_1}^+ a_{k_2} - a_{k_1}^+ a_{k_2} a_k^+ a_{k'}$$

$$a_k^+ \left(\delta_{k'k_1} - a_{k_1}^+ a_{k'}\right) a_{k_2} = \delta_{k'k_1}\, a_k^+ a_{k_2} - a_k^+ a_{k_1}^+ a_{k'} a_{k_2}$$

$$= \delta_{k'k_1}\, a_k^+ a_{k_2} - a_{k_1}^+ a_k^+ a_{k_2} a_{k'} =$$

$$= \delta_{k'k_1}\, a^+_k a_{k_2} - a^+_{k_1}\left(\delta_{kk_2} - a_{k_2} a^+_k\right) a_{k'} =$$

$$= \delta_{k'k_1}\, a^+_k a_{k_2} - \delta_{kk_2}\, a^+_{k_1} a_{k'} + a^+_{k_1} a_{k_2} a^+_k a_{k'}$$

Thus $\langle [\,,\,] \rangle = \underbrace{\langle a^+_k a_{k_2}\rangle}_{\langle a^+_k a_k\rangle \delta_{kk_2}} \delta_{k'k_1} - \underbrace{\langle a^+_{k_1} a_{k'}\rangle}_{\langle a^+_{k'} a_{k'}\rangle \delta_{k_1k'}} \delta_{kk_2}$

$$\chi_0(\vec{r},t) = -i\,\theta(t)\,\frac{1}{V^2}\sum_{\substack{kk'\\ k_1k_2}} \left(n_k\,\delta_{kk_2}\delta_{k'k_1} - n_{k'}\,\delta_{k'k_1}\delta_{kk_2}\right)$$

$$\cdot\, e^{-i(\vec{k}-\vec{k}')\cdot\vec{r}}\, e^{i(\varepsilon_k-\mu)t/\hbar}\, e^{-i(\varepsilon_{k'}-\mu)t/\hbar}$$

Now recall that

$$\chi(\vec{q},\omega) = \int d\vec{r}\, dt\, e^{-i\vec{q}\cdot\vec{r}+i\omega t}\,\chi(\vec{r},t)$$

So that two integrals are needed

$$\int d\vec{r}\, e^{-i\vec{q}\cdot\vec{r}}\, e^{-i(\vec{k}-\vec{k}')\cdot\vec{r}} = \int d\vec{r}\, e^{-i\vec{r}\cdot(\vec{q}+\vec{k}-\vec{k}')}$$

$$= V\cdot\delta_{\vec{q}+\vec{k}-\vec{k}',0} \quad \Rightarrow \vec{k}' = \vec{k}+\vec{q}$$ [here we think of finite volume system

$$\int_{-\infty}^{\infty} dt\,\left(-i\,\theta(t)\right) e^{it\left(\frac{\varepsilon_k-\mu}{\hbar} - \frac{\varepsilon_{k'}-\mu}{\hbar}\right)} e^{i\omega t} =$$

$$= -i\int_0^{\infty} dt\, e^{it\left(\omega+i\delta+\frac{\varepsilon_k-\varepsilon_{k'}}{\hbar}\right)} =$$ ($\delta = +0$ comes from $\theta(t)$)

$$= (-i) \quad \frac{e^{it\left(\omega + i\delta + \frac{\varepsilon_k - \varepsilon_{k'}}{\hbar}\right)}\Big|_0^\infty}{i\left(\omega + i\delta + \frac{\varepsilon_k - \varepsilon_{k'}}{\hbar}\right)} = (-i)\,\frac{-1}{i\left(\omega + \frac{\varepsilon_k - \varepsilon_{k'}}{\hbar} + i\delta\right)}$$

Collect together

$$\chi(\vec{q},\omega) = \frac{1}{V}\sum_{\vec{k}} \frac{n_{\vec{k}} - n_{\vec{k}+\vec{q}}}{\omega + i\delta + \frac{1}{\hbar}\left(\varepsilon_{\vec{k}} - \varepsilon_{\vec{k}+\vec{q}}\right)}$$

$$n_{\vec{k}} \equiv \langle \hat{a}^+_k \hat{a}_k \rangle$$

$$\varepsilon_{\vec{k}} = \begin{cases} \frac{\hbar^2 \vec{k}^2}{2m} & \text{for free system of fermions;} \\ \text{can be anything} \rightarrow \text{we have not used its explicit form yet!} \end{cases}$$

Moreover, $\varepsilon_k - \varepsilon_{k+q} = (\varepsilon_k - \mu) - (\varepsilon_{k+q} - \mu)$.

(And we set $\hbar = 1$ now — it can be restored at any stage if necessary)

Static susceptibility, $\omega = 0$:

$$\chi(\vec{q},0) = \frac{1}{V}\sum_k \frac{n_k - n_{k+q}}{i\delta + \varepsilon_k - \varepsilon_{k+q}}$$

Using $\mathrm{Im}\left(\frac{1}{x+i\delta}\right) = \mathrm{Im}\left(\frac{P}{x} - i\pi\delta(x)\right) = -\pi\delta(x)$

$$\mathrm{Im}\,\chi(\vec{q},0) = -\pi\frac{1}{V}\sum_k (n_k - n_{k+q})\,\delta(\varepsilon_k - \varepsilon_{k+q})$$

But $n_k = \langle a_k^+ a_k\rangle = f(\varepsilon_k) = \frac{1}{e^{\frac{\varepsilon_k-\mu}{k_BT}}-1}$ (Fermi-Dirac dist.)

$$\mathrm{Im}\,\chi(\vec{q},0) = -\pi\frac{1}{V}\sum_k \left(f(\varepsilon_k) - f(\varepsilon_{k+q})\right)\delta(\varepsilon_k - \varepsilon_{k+q})$$

$\delta(\varepsilon_k - \varepsilon_{k+q})$ requires $\varepsilon_k = \varepsilon_{k+q}$ which makes the two Fermi functions identical, $f(\varepsilon_k) = f(\varepsilon_{k+q}) \Rightarrow \underline{\mathrm{Im}\,\chi(\vec{q},0) = 0}$

Static susceptibility is real — quite obvious statement as there can not be energy absorption from the static field.

$$\chi(\vec{q},0) = \mathrm{Re}\left(\chi(\vec{q},0)\right) = \frac{1}{V}\sum_k \frac{n_k - n_{k+q}}{\varepsilon_k - \varepsilon_{k+q}}$$

Consider now uniform $\vec{q}\to 0$ limit: convenient to write everything in terms of $\xi_k = \varepsilon_k - \mu$

$$\chi(\vec{q}\to 0, \omega=0) = \frac{1}{V}\sum_k \frac{f(\xi_k) - f(\xi_{k+q})}{\xi_k - \xi_{k+q}} =$$

$$= \frac{1}{V} \sum_k \frac{f(\xi_k) - f(\xi_k + \xi_{k+q} - \xi_k)}{\xi_k - \xi_{k+q}} =$$

$$= \frac{1}{V} \sum_k \frac{f(\xi_k) - \left[f(\xi_k) + (\xi_{k+q} - \xi_k) \frac{\partial f}{\partial \xi_k} \right]}{\xi_k - \xi_{k+q}} =$$

$$= \frac{1}{V} \sum_k \frac{\partial f(\xi_k)}{\partial \xi_k} = \int d\xi \, \underbrace{\mathcal{N}(\xi)}_{\text{density of states}} \frac{\partial f}{\partial \xi} = -\mathcal{N}(0)$$

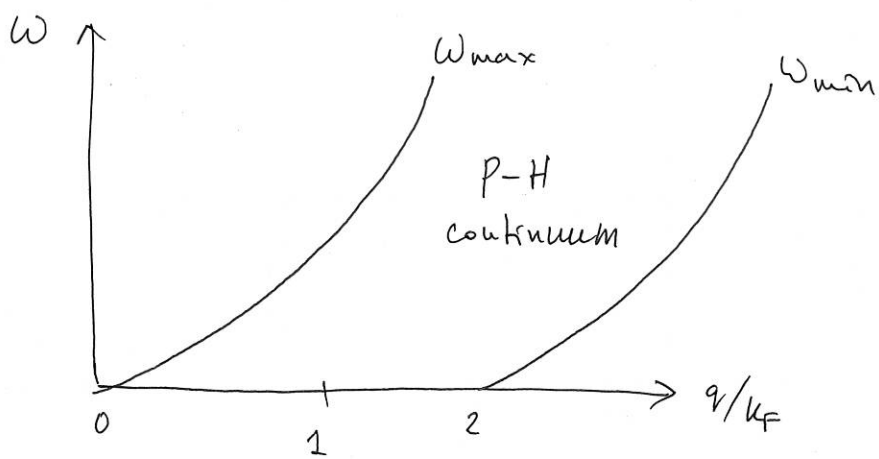
Density of states at the Fermi level.

Since $\frac{\partial f}{\partial \xi} \xrightarrow[T \to 0]{} -\delta(\xi)$ (this is important property of Fermi distribution function)

$$\mathrm{Im}\,\chi(\vec{q},\omega) = -\frac{\pi}{V}\sum_k \Big(f(\xi_k) - f(\xi_{k+q})\Big)\,\delta(\omega + \xi_k - \xi_{k+q})$$

$$= -\frac{\pi}{V}\sum_k \Big(f(\xi_k) - f(\omega + \xi_k)\Big)\,\delta\left(\omega - \frac{\vec{k}\cdot\vec{q}}{m} - \frac{\vec{q}^{\,2}}{2m}\right)$$

for $\xi_k = \dfrac{\vec{k}^2 - k_F^2}{2m}$



$$\omega_{min} = -\frac{kq}{m} + \frac{q^2}{2m} \quad \left(\text{for } \cos\theta_{k,q} = -1\right) > 0$$

$$\Rightarrow \frac{q^2}{2m} > \frac{kq}{m} \quad \Rightarrow q > 2k$$

$\underline{d=3}$ $\quad \operatorname{Im}\chi(\vec{q},\omega) = -\pi \int \frac{d\vec{k}}{(2\pi)^3} \left(f(\xi_k) - f(\xi_{k+q})\right) \delta(\omega + \xi_k - \xi_{k+q})$

$$= -\pi \int \frac{d\vec{k}}{(2\pi)^3} \left(f(\xi_k) - f(\omega + \xi_k)\right) \delta\left(\omega - \frac{kq\cos\theta}{m} - \frac{q^2}{2m}\right)$$

$$\int \frac{d^3\vec{k}}{(2\pi)^2} = \int d\xi \, \frac{k_F^2}{v_F} \int_0^{2\pi} \frac{d\varphi}{2\pi} \int_0^{\pi} \frac{d\theta \, \sin\theta}{(2\pi)^2}$$

$\xi_k = v_F k \;,\; v_F = \frac{k_F}{m}$ $\qquad$ θ = angle between $\vec{k} = \vec{k}_F$ and $\vec{q}$

$$\operatorname{Im}\chi = -\frac{\pi}{(2\pi)^2} (m k_F) \int_0^{\pi} d\theta \sin\theta \int_{-\infty}^{\infty} d\xi \left(f(\xi) - f(\xi+\omega)\right) \cdot$$
$$\cdot \delta\left(\omega - k_F q\cos\theta - \frac{q^2}{2m}\right)$$

$$\underbrace{\int_0^{\pi} d\theta \sin\theta}_{\int_{-1}^{1} d\cos\theta} \delta\left(\omega - v_F q\cos\theta - \frac{q^2}{2m}\right) = \int_{-1}^{1} dz \, \delta\left(\omega - \frac{q^2}{2m} - v_F q z\right) =$$

$\cos\theta = z$

$$= \frac{1}{v_F q} \int_{-v_F q}^{v_F q} d\tilde{z} \, \delta\left(\omega - \frac{q^2}{2m} - \tilde{z}\right)$$

$$= \frac{1}{v_F q} \theta\left(1 - \frac{\left|\omega - \frac{q^2}{2m}\right|}{v_F q}\right).$$

$$-v_F q < \omega - \frac{q^2}{2m} < v_F q$$

$$\left|\omega - \frac{q^2}{2m}\right| < v_F q$$

$$\int_{-\infty}^{\infty} d\xi \left(f(\xi) - f(\xi+\omega)\right) = (T=0) = \int_{-\infty}^{0} d\xi - \int_{-\infty}^{-\omega} d\xi = \int_{-\omega}^{0} d\xi = \omega.$$

$f(\xi) = \theta(-\xi)$

The same integral at finite T

$\underline{T \neq 0}$ $I = \int d\xi\, f(\xi) = \int_{-\infty}^{\infty} \frac{d\xi}{e^{\frac{\xi}{T}}+1} = T \int_{-\infty}^{\infty} dx \frac{1}{e^{x}(1+e^{-x})} =$

$$= T \int_{-\infty}^{\infty} dx \frac{e^{-x}}{1+e^{-x}} = -T \int_{-\infty}^{\infty} d(e^{-x}) \frac{1}{1+e^{-x}} = -T \ln\left(1+e^{-x}\right)\Big|_{-\infty}^{\infty}$$

In fact the lower limit of ξ-integration is not $(-\infty)$ but rather $(-E_F)$, since $\xi_k = \frac{k^2}{2m} - \mu = \frac{k^2}{2m} - E_F$

$$\Rightarrow I = -T \ln\left(1+e^{-x}\right)\Big|_{-\frac{E_F}{T}}^{\infty} = -T\left(\ln 1 - \ln\left(1+e^{\frac{E_F}{T}}\right)\right)$$

$$= T \ln\left(1+e^{E_F/T}\right) = T \ln e^{\frac{E_F}{T}}\left(1+e^{-E_F/T}\right) = T\left\{\frac{E_F}{T} + \ln\left(1+e^{-\frac{E_F}{T}}\right)\right\}$$

$$= E_F + T \cdot O\left(e^{-E_F/T}\right).$$

Then $\int_{-E_F}^{\infty} d\xi \left(f(\xi) - f(\xi+\omega)\right) = \int_{-E_F}^{\infty} d\xi\, f(\xi) - \int_{-E_F+\omega}^{\infty} d\xi'\, f(\xi')$

$$= E_F - (E_F - \omega) = \omega$$ up to exponentially small in $e^{-E_F/T}$ terms!

$$\Rightarrow \operatorname{Im}\chi(\vec{q},\omega) = -\frac{1}{2}\pi \underbrace{\left(\frac{m k_F}{2\pi^2}\right)}_{} \frac{\omega}{v_F q}\, \theta\left(1 - \frac{\left|\omega - \frac{q^2}{2m}\right|}{v_F q}\right).$$

ν_0 = density of states at the Fermi level.

Next use Kramers–Kronig:

$$\chi'(\vec{q},\Omega) = \frac{1}{\pi} P \int_{-\infty}^{\infty} d\omega \frac{\chi''(\vec{q},\omega)}{\omega - \Omega} \quad ; \quad \chi' = \operatorname{Re}\chi, \quad \chi'' = \operatorname{Im}\chi$$

$$\chi''(\vec{q},\Omega) = -\frac{1}{\pi} P \int_{-\infty}^{\infty} d\omega \frac{\chi'(\vec{q},\omega)}{\omega - \Omega}$$

$$\chi'(\vec{q},\Omega) = \frac{1}{\pi} P \int_{-\infty}^{\infty} d\omega \left(\frac{-\nu_0 \pi}{2 v_F q}\right) \frac{\omega\, \theta\left(1 - \frac{\left|\omega - \frac{q^2}{2m}\right|}{v_F q}\right)}{\omega - \Omega}$$

$$-v_F q < \omega - \frac{q^2}{2m} < v_F q$$

$$-v_F q + \frac{q^2}{2m} < \omega < v_F q + \frac{q^2}{2m}$$

$$= -\frac{\nu_0}{2 v_F q} P \int_{-v_F q + \frac{q^2}{2m}}^{v_F q + \frac{q^2}{2m}} d\omega \underbrace{\frac{\omega}{\omega - \Omega}}_{1 + \frac{\Omega}{\omega - \Omega}} = -\frac{\nu_0}{2 v_F q}\left(\omega + \Omega \ln|\omega - \Omega|\right)\Bigg|_{-v_F q + \frac{q^2}{2m}}^{v_F q + \frac{q^2}{2m}}$$

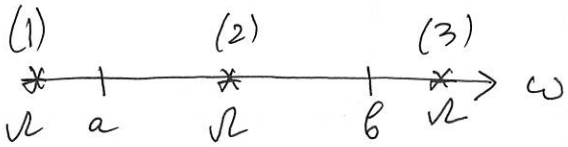
$$= -\frac{\nu_0}{2 v_F q}\left(2 v_F q + \Omega \ln\left|\frac{v_F q + \frac{q^2}{2m} - \Omega}{-v_F q + \frac{q^2}{2m} - \Omega}\right|\right) =$$

$$= -\nu_0 \left\{1 + \frac{\Omega}{2 v_F q} \ln\left|\frac{\Omega - v_F q - \frac{q^2}{2m}}{\Omega + v_F q - \frac{q^2}{2m}}\right|\right\}.$$

Comment on importance of Principal Value integral:

We are looking to calculate $J = \text{P.V.} \int_b^a d\omega \, \frac{1}{\omega - \Omega}$

$a = v_F q + \frac{q^2}{2m}$, $b = -v_F q + \frac{q^2}{2m}$.



Have to distinguish 3 situations:

(1) $\Omega < a < b$

(2) $a < \Omega < b$, (3) $a < b < \Omega$

(1) $$J_1 = \text{P.V.} \int_a^b \frac{d\omega}{\omega - \Omega} = \ln(\omega - \Omega)\Big|_a^b = \ln \frac{b-\Omega}{a-\Omega} = \ln\left|\frac{b-\Omega}{a-\Omega}\right|$$

because $b - \Omega > a - \Omega > 0$ by assumption here.

(2) $$J_2 = \int_a^{\Omega-\varepsilon} d\omega \, \frac{1}{\omega-\Omega} + \int_{\Omega+\varepsilon}^b d\omega \, \frac{1}{\omega - \Omega} = \ln(\omega-\Omega)\Big|_a^{\Omega-\varepsilon} + \ln(\omega-\Omega)\Big|_{\Omega+\varepsilon}^b =$$

$$= \ln \frac{\Omega - \varepsilon - \Omega}{a - \Omega} + \ln \frac{b-\Omega}{\Omega+\varepsilon-\Omega} =$$

$$= \ln \frac{-\varepsilon}{a-\Omega} + \ln \frac{b-\Omega}{\varepsilon} = \ln \frac{\varepsilon}{\Omega - a} + \ln \frac{b-\Omega}{\varepsilon} =$$

$$= \ln \frac{b-\Omega}{\Omega - a} = \ln\left|\frac{b-\Omega}{a-\Omega}\right|.$$ because here $|a - \Omega| = \Omega - a$.

(3) $$J_3 = \int_a^b \frac{d\omega}{\omega - \Omega} = \ln(\omega - \Omega)\Big|_a^b = \ln \frac{b-\Omega}{a-\Omega} = \ln \frac{\Omega - b}{\Omega - a} =$$

$$= \ln\left|\frac{b-\Omega}{a-\Omega}\right|.$$

Thus P.V. prescription guarantees that

$J = \ln\left|\frac{b-\Omega}{a-\Omega}\right|$ is given by <u>real</u> function $\ln|x|$, and not $\ln(x)$.